

Model Lab

Every prior, same slate — 2026-09-30

SERVED PRIOR

Served (full season)

PRIORS COMPARED

11

HELD-OUT PA

37,035

Held-Out Scores

Fitted on plate appearances before 2026-08-22, scored on the 37,035 that came after (actual HR rate 3.02%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5849	1.57×	0.02924	0.13454	0.00494
KNN comps Prior is the pooled rate of the nearest comparables.	0.5936	1.51×	0.02923	0.13400	0.00572
SVR Support-vector regression on the same feature matrix.	0.5905	1.46×	0.02930	0.13505	0.00744
Random forest Forest regression; a learned similarity metric.	0.5912	1.47×	0.02928	0.13447	0.00651
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5936	1.51×	0.02923	0.13400	0.00572
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5936	1.51×	0.02923	0.13400	0.00572
Linear regression Ordinary least squares on the same features.	0.5922	1.51×	0.02925	0.13439	0.00514
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5935	1.51×	0.02929	0.13449	0.00709
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5923	1.49×	0.02926	0.13430	0.00607
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5908	1.45×	0.02930	0.13470	0.00759

Blend weights (forest variant): KNN 1.00, SVR 0.00, RF 0.00

Shrinkage K per prior: league 258, knn 258, svr 258, rf 258, core 258, ensemble 258, linear 258, logistic 258, nn 258, xgb 258 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (4,331 PA, actual HR rate 2.47%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5392	1.40×	0.02412	0.11617	0.00831
KNN comps Prior is the pooled rate of the nearest comparables.	0.5615	1.22×	0.02424	0.11767	0.01599
SVR Support-vector regression on the same feature matrix.	0.5680	1.12×	0.02469	0.12456	0.02510
Random forest Forest regression; a learned similarity metric.	0.5743	1.12×	0.02456	0.12061	0.02294
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5615	1.22×	0.02424	0.11767	0.01599
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5615	1.22×	0.02424	0.11767	0.01599
Linear regression Ordinary least squares on the same features.	0.5656	1.12×	0.02442	0.12089	0.01983
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5713	0.93×	0.02457	0.12034	0.02272
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5710	1.12×	0.02436	0.11901	0.01827
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5804	1.12×	0.02473	0.12183	0.02548

Slate Agreement

Served (full season) · 6/6 shared 1/6 hit

Fernando Tatis Jr.
Ben Rice
Kyle Schwarber ✓
Ronald Acuña Jr.
Munetaka Murakami
Pete Crow-Armstrong

KNN comps · 6/6 shared 1/6 hit

Fernando Tatis Jr.
Ben Rice
Kyle Schwarber ✓
Ronald Acuña Jr.
Munetaka Murakami
Pete Crow-Armstrong

Random forest · 6/6 shared 1/6 hit

Ben Rice
Fernando Tatis Jr.
Kyle Schwarber ✓
Ronald Acuña Jr.
Pete Crow-Armstrong
Munetaka Murakami

Forest blend · 6/6 shared 1/6 hit

Fernando Tatis Jr.
Ben Rice
Kyle Schwarber ✓
Ronald Acuña Jr.
Munetaka Murakami
Pete Crow-Armstrong

Logistic regression · 5/6 shared 1/6 hit

Flat league · 5/6 shared 1/6 hit

19.4%	Fernando Tatis Jr.	19.1%
19.2%	Ben Rice	17.9%
18.7%	Kyle Schwarber ✓	17.2%
17.8%	Ronald Acuña Jr.	17.2%
17.4%	Pete Crow-Armstrong	16.5%
17.3%	Manny Machado	16.0%

SVR · 5/6 shared 1/6 hit

19.6%	Ben Rice	19.7%
19.2%	Kyle Schwarber ✓	19.3%
18.7%	Fernando Tatis Jr.	19.1%
17.9%	Ronald Acuña Jr.	18.1%
17.3%	Pete Crow-Armstrong	18.0%
17.3%	Yordan Alvarez	17.8%

Core (KNN+SVR) · 6/6 shared 1/6 hit

19.6%	Fernando Tatis Jr.	19.6%
19.2%	Ben Rice	19.2%
19.1%	Kyle Schwarber ✓	18.7%
18.0%	Ronald Acuña Jr.	17.9%
17.8%	Munetaka Murakami	17.3%
17.6%	Pete Crow-Armstrong	17.3%

Linear regression · 5/6 shared 1/6 hit

19.6%	Fernando Tatis Jr.	19.6%
19.2%	Ben Rice	19.2%
18.7%	Kyle Schwarber ✓	19.0%
17.9%	Ronald Acuña Jr.	17.9%
17.3%	Pete Crow-Armstrong	17.4%
17.3%	Yordan Alvarez	17.2%

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
19. Ty France San Diego Padres · vs RHP · 21 HR / 504 PA	14.1%	13.6%	14.1%	14.1%	14.0%	14.1%	14.1%	14.1%	14.0%
20. Jahmai Jones Boston Red Sox · vs LHP · 6 HR / 181 PA	13.8%	12.1%	13.7%	13.5%	13.7%	13.7%	13.7%	13.8%	13.9%
21. Lucas Spence Houston Astros · vs RHP · 4 HR / 67 PA	13.6%	12.1%	13.7%	12.7%	14.0%	13.7%	13.7%	13.8%	13.4%
22. Alex Bregman Chicago Cubs · vs RHP · 28 HR / 706 PA	13.3%	13.4%	13.2%	13.6%	13.5%	13.2%	13.2%	13.6%	13.6%
23. Seiya Suzuki Chicago Cubs · vs RHP · 26 HR / 638 PA	13.2%	12.8%	13.2%	13.5%	13.5%	13.2%	13.2%	13.2%	13.4%
24. Alec Bohm Philadelphia Phillies · vs RHP · 15 HR / 633 PA	12.9%	13.1%	12.5%	12.2%	12.5%	12.5%	12.5%	12.2%	12.1%
25. Wilyer Abreu Boston Red Sox · vs LHP · 24 HR / 684 PA	12.9%	11.8%	12.6%	12.9%	12.9%	12.6%	12.6%	12.7%	13.0%
26. Willson Contreras Boston Red Sox · vs LHP · 27 HR / 540 PA	12.7%	11.2%	12.8%	12.8%	12.8%	12.8%	12.8%	13.0%	13.6%
27. Edmundo Sosa Philadelphia Phillies · vs RHP · 9 HR / 252 PA	12.6%	12.5%	12.3%	12.3%	12.3%	12.3%	12.3%	12.3%	12.4%
28. Andrew Benintendi Chicago White Sox · vs RHP · 15 HR / 470 PA	12.6%	11.5%	12.4%	12.1%	12.1%	12.4%	12.4%	12.6%	12.7%
29. Trea Turner Philadelphia Phillies · vs RHP · 19 HR / 704 PA	12.1%	13.1%	12.3%	12.8%	12.6%	12.3%	12.3%	12.5%	12.5%
30. Cody Bellinger New York Yankees · vs RHP · 17 HR / 566 PA	12.1%	12.3%	11.9%	11.8%	12.0%	11.9%	11.9%	12.0%	12.0%
31. Trent Grisham New York Yankees · vs RHP · 18 HR / 485 PA	12.0%	11.0%	12.0%	11.8%	11.6%	12.0%	12.0%	11.5%	11.5%
32. Brandon Marsh Philadelphia Phillies · vs RHP · 18 HR / 569 PA	11.9%	11.4%	12.1%	11.9%	11.9%	12.1%	12.1%	12.0%	11.9%
33. Christian Walker	11.9%	11.9%	11.7%	12.3%	11.9%	11.7%	11.7%	12.0%	12.2%

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
Houston Astros · vs RHP · 29 HR / 639 PA									
34. Yainer Diaz Houston Astros · vs RHP · 14 HR / 365 PA	11.7%	11.9%	11.3%	12.1%	11.5%	11.3%	11.3%	11.8%	11.8%
35. Xander Bogaerts San Diego Padres · vs RHP · 14 HR / 610 PA	11.6%	11.7%	11.4%	11.3%	11.5%	11.4%	11.4%	11.7%	11.3%
36. Jazz Chisholm Jr. New York Yankees · vs RHP · 19 HR / 532 PA	11.4%	10.9%	11.6%	11.0%	11.1%	11.6%	11.6%	11.2%	11.1%
37. Colson Montgomery Chicago White Sox · vs RHP · 32 HR / 642 PA	11.4%	10.2%	11.5%	11.1%	11.3%	11.5%	11.5%	11.6%	12.1%
38. Ethan Salas San Diego Padres · vs RHP · 2 HR / 40 PA	11.4%	11.5%	11.3%	12.2%	11.1%	11.3%	11.3%	11.5%	10.8%
39. Isaac Paredes Houston Astros · vs RHP · 25 HR / 663 PA	11.3%	11.7%	11.2%	11.3%	11.1%	11.2%	11.2%	11.3%	11.3%
40. Austin Wells New York Yankees · vs RHP · 14 HR / 392 PA	11.2%	10.6%	11.2%	10.6%	10.7%	11.2%	11.2%	10.8%	10.6%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.